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## AWS CloudWatch Logs Agent vs. Unified CloudWatch Agent

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### ♦ 1. CloudWatch Logs Agent (Legacy)

#### Purpose:

- **Sends log files** from EC2 to **CloudWatch Logs**
- No support for custom metrics, CPU/memory/disk, etc.

#### Architecture:

- Installed manually on EC2 (Linux)
- Uses `/etc/awslogs/awslogs.conf` for configuration
- Sends logs to CloudWatch Logs **only**

#### Common Use Cases:

- Send **Apache/Nginx logs**, **custom app logs**, etc.
- Lightweight log shipping solution

#### Limitations:

- No support for **metrics**
- No **Windows** support

- Being **replaced by Unified Agent**
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## ♦ 2. Unified CloudWatch Agent (Recommended)

### ✓ Purpose:

- Collects both:
  - **System-level metrics**: CPU, memory, disk, network
  - **Application logs**

### 💡 Features:

- **Cross-platform** (Linux & Windows)
- Supports EC2, on-prem, hybrid, and container-based systems
- Can push **custom metrics** and **log files**
- Configurable via **SSM Parameter Store** or local JSON

### 🏗️ Architecture:

- Installed from AWS repo or manually
- Configurable via:
  - [amazon-cloudwatch-agent-config-wizard](#)
  - or a pre-built JSON config file

## Common Use Cases:

- Monitor:
    - **Memory usage** (not available in native EC2 metrics)
    - **Disk space, network IO**
    - Application logs
  - Use in **hybrid environments** and **centralized monitoring**
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## Quick Comparison

| Feature                | CloudWatch Logs Agent | Unified CloudWatch Agent     |
|------------------------|-----------------------|------------------------------|
| Collect Logs           | ✓ Yes                 | ✓ Yes                        |
| Collect Metrics        | ✗ No                  | ✓ Yes (CPU, Mem, Disk, etc.) |
| Supports Windows       | ✗ No                  | ✓ Yes                        |
| Configurable via SSM   | ✗ No                  | ✓ Yes                        |
| Custom Metrics Support | ✗ No                  | ✓ Yes                        |
| Recommended by AWS     | ⚠ Legacy              | ✓ Yes (recommended)          |

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## Deployment Overview

### ✓ Installing Unified CloudWatch Agent (Amazon Linux 2):

```
sudo yum install amazon-cloudwatch-agent -y
sudo
/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard
```

Then start the agent:

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
-a fetch-config -m ec2 \
-c file:/opt/aws/amazon-cloudwatch-agent/config.json -s
```

### ✓ IAM Role Permission Required:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "cloudwatch:PutMetricData",
        "logs:PutLogEvents",
        "logs:CreateLogStream",
        "logs:CreateLogGroup",
        "ssm:GetParameter"
      ],
      "Resource": "*"
    }
  ]
}
```

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### 🧠 Exam & Architect Tips (SAA-C03):

- Use **Unified Agent** when you need **memory or disk metrics**
  - Required for **custom metrics**
  - Use **SSM-based configuration** for fleet-wide deployment
  - Legacy **Logs Agent** is not supported on Windows
  - **Alarms** can only be created on metrics — **Unified Agent enables this**
  - Use **CloudFormation**, **EC2 user data**, or **SSM State Manager** to automate deployment
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